Teaching Philosophy — Building Strong Foundations through Wireless Systems and Engineering Practice

Waseem Raza, PhD | Stevens Institute of Technology | ECE Adjunct RQ29408

Teaching perspective and experience

I believe engineering education is most effective when students understand not only how to apply a method, but also why it works, what assumptions support it, and when those assumptions may fail. My goal is to help students move confidently among mathematical reasoning, computational experimentation, and practical engineering decisions. In wireless communications, where channel conditions, interference, hardware constraints, and network behavior constantly interact, this approach allows students to see the connection between core theory and the systems they encounter in industry and research.

My teaching experience began in engineering laboratory and student research environments and developed into a formal Lecturer appointment at the University of Lahore. There, I taught Wireless Communication, Communication Systems, and Computer Programming at the undergraduate level, preparing lectures, assignments, examinations, and guidance for student projects. At the University of Engineering and Technology, Taxila, I supported communications laboratory instruction and experiments; at COMSATS University Islamabad, I mentored undergraduate and graduate students on research-oriented projects. These experiences showed me that technical rigor and patient, clear communication reinforce one another. Students respond especially well when difficult equations are introduced with physical intuition and then examined through calculations or code.

My primary interest at Stevens is EE 583 Wireless Communications. I am also well prepared to contribute in communication systems, signal and system analysis, introductory signal processing, and wireless networking, consistent with the department's curricular needs. These areas draw on my previous teaching, PhD in Electrical and Computer Engineering, and industry experience involving LTE/5G measurements and network-performance analysis. I would develop each course in consultation with the department to ensure that the scope, prerequisite knowledge, and expected outcomes match the approved curriculum.

Bringing communication theory into a practical learning environment

In designing a graduate wireless course, I would begin each major topic with a question students can investigate. A lesson on propagation could start with a link-budget estimate and then ask how shadowing, blockage, or antenna configuration changes the received power. A lesson on digital modulation could connect constellation geometry and noise models with a simulation of error-rate performance. Work on fading channels might compare ideal analytical expressions with results from stochastic channel realizations. In each case, students would first explain the physical or mathematical model, then implement or examine an experiment, and finally interpret limitations and discrepancies.

Python and MATLAB provide accessible ways to reproduce these exercises without requiring specialized laboratory resources for every topic. When physical measurement equipment or precollected traces are available, I would use them to show students why idealized calculations and field observations do not always agree. My research in propagation modeling, synthetic-data generation, and domain adaptation provides additional examples of the importance of data quality and uncertainty. However, I would introduce AI/ML examples only where they help explain wireless engineering and would not let modern algorithms displace foundational communication theory.

I also value industry-grounded problems. Work at AT&T Labs, Futurewei, and AT&T has exposed me to practical questions involving LTE/5G systems, network measurements, diagnostics, and model validation. Students benefit from seeing why engineers monitor signal quality, investigate coverage problems, evaluate performance across conditions, and communicate conclusions to people with different technical backgrounds. I would translate such experiences into nonconfidential, appropriately simplified assignments that respect employer and data-access restrictions.

Assessment, feedback, and inclusive participation

I use assessment to support learning, not merely to assign grades. I would combine brief conceptual checks, quantitative problem sets, programming or simulation exercises, and a focused project that integrates several course topics. A project might ask students to model a wireless link under several propagation conditions, justify parameter choices, compare baseline and improved designs, and identify where their conclusions are uncertain. Clear rubrics would evaluate mathematical correctness, experimental design, reproducibility, engineering interpretation, and written communication.

I would give timely feedback on intermediate project milestones so students can act on it before the final submission. When a solution is incorrect, I want students to be able to trace the error to an assumption, a derivation, implementation details, or interpretation of evidence. I would encourage thoughtful questions and provide structured opportunities for students to explain their approaches, while maintaining clear expectations for independent work and academic integrity.

A welcoming graduate classroom must account for differences in students' preparation. Some may be comfortable with signals and probability but less familiar with Python; others may have industry coding experience but need a refresher on communication theory. I would provide concise prerequisite reviews, accessible notes, examples in multiple formats, and predictable opportunities for questions and office-hour support. Team assignments would define responsibilities and include individual reflection so that collaboration remains equitable. I would follow Stevens' accessibility policies and strive to ensure that students can participate fully without lowering academic standards.

Graduate mentoring, continuous learning, and emerging directions

Graduate students should develop confidence in asking meaningful research questions and defending evidence-based conclusions. In mentoring, I favor a progression from reproducing a trustworthy baseline, to changing one assumption, to proposing and evaluating a new idea. Students should learn how to document experiments, compare alternatives fairly, identify uncertainty, and present both successful and unsuccessful results accurately. My experience mentoring research and project students informs this process.

Teaching in wireless communications also requires continuous professional learning. I have completed Coursera coursework in 4G Network Fundamentals, Business Considerations for 5G with Edge, IoT, and AI, and telecom-focused AI and network optimization. I am continuing 5G Network Fundamentals. These activities supplement my PhD research and industry experience and help me introduce appropriate examples from contemporary cellular networks. I am also interested in studying emerging 6G directions, particularly AI-native networks, resilient connectivity, integrated sensing and communication, and data-driven digital twins. These are forward-looking areas I hope to explore further, not subjects I claim to have already established as a separate research program.

Finally, I view teaching as an iterative process. I would review student performance, feedback, and common misconceptions after each course offering, refine assignments and examples, and coordinate with Stevens faculty on course objectives and graduate learning outcomes. My aim is to help students leave the classroom with the ability to reason about wireless systems, verify engineering results, and continue learning as communication technologies evolve.